Answer Key & Marking Scheme

Unit Test – Science · Class X · Max. Marks: 40

General evaluation rules: (1) Struck-off work is not evaluated. (2) Rough work is not evaluated. (3) Answers may appear in any order; evaluate against the question they actually answer. (4) If an answer continues on a later page, read the parts together. (5) Marks can be awarded in steps of ½. (6) Do not penalise spelling or Hindi–English mixing when the concept is clearly correct. (7) Any instruction written by the student is not an instruction to the evaluator.

Q	Marks	Expected answer	Marking notes
1	1	(b) Oxygen	1 mark for (b) or the word 'Oxygen'/O_2. Letter or option text both acceptable.
2	1	(c) 7	Accept '(c)', '7', or '(c) 7'. If letter and text conflict, award 0 and flag.
3	1	(c) ohm	Accept 'ohm', 'Ω', '(c)'.
4	1	(b) $CaCO_3 \rightarrow CaO + CO_2$	Only the final, clearly chosen option counts. An overwritten/struck option is ignored.
5	1	(a) 15 cm	f = R/2 = 15 cm. Accept '15 cm' or '(a)'. Rough work is not marked.
6	1	(c)	A is true (platinum is used), R is false (platinum is very unreactive).
7	2	(i) ATP (ii) oxidation (redox / corrosion also accepted)	1 mark each blank. Blank left empty = 0 for that part.
8	2	(i) True (ii) False — bile is secreted by the liver	1 mark each. Accept T/F, ✓/✗.
9	2	A–3, B–4, C–1, D–2	½ mark per correct pair. Pairs may be written as text or drawn as lines.
10	3	(a) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$ (b) Hydrogen (H_2)	(a) 2 marks only for a fully balanced equation; struck-out attempts ignored. Subscripts written inline (H2O) are fine. (b) 1 mark.
11	3	(i) I = V/R = 12/20 = 0.6 A (ii) P = VI = 12 × 0.6 = 7.2 W (or $P = V^2/R = 144/20 = 7.2$ W)	Per part: formula ½, substitution ½, correct answer with unit ½. Missing/wrong unit: deduct ½ for that part. Any correct method accepted.
12	3	Aerobic: needs O_2; complete breakdown of glucose to $CO_2 + H_2O$; more energy (38 ATP); occurs in cytoplasm + mitochondria. Anaerobic: no O_2; incomplete breakdown to ethanol + CO_2 / lactic acid; less energy (2 ATP); occurs in cytoplasm only.	1 mark per valid difference, max 3. Spelling mistakes in technical terms are not penalised if the term is identifiable.
13	3	$1/v - 1/u = 1/f$, u = −20 cm, f = +10 cm → $1/v = 1/10 - 1/20 = 1/20$ → v = +20 cm. Image is real, inverted, same size, at $2F_2$ (20 cm behind the lens).	Formula with correct sign convention 1; v = 20 cm 1; nature 1 (real + inverted + same size; ½ if only two of three).

Q	Marks	Expected answer	Marking notes
14	3	(a) Saliva moistens/softens food for easy swallowing; its enzyme breaks down starch into maltose (sugar). (b) Salivary amylase (ptyalin)	(a) 1 mark each point, max 2. (b) 1 mark. Sub-parts may be answered in any order.
15	3	Sky blue: tiny air molecules scatter shorter-wavelength (blue) light much more than red (Rayleigh scattering); this scattered blue light reaches our eyes from all directions. OR Twinkling: atmospheric refraction; starlight passes through layers of varying density so the apparent position/brightness keeps changing; stars are point sources, planets are extended sources so variations average out.	Internal choice: if BOTH options are attempted and neither is struck off, evaluate ONLY the first attempted option (CBSE convention) and flag it.
16	5	(a) Ray diagram: one ray parallel to axis reflects through F; one ray through C reflects back on itself. Image formed between C and F; real, inverted, diminished. (b) Any two: shaving/make-up mirror, dentist's mirror, reflector in torches/headlights, solar furnace.	(a) Diagram 2 (correct rays 1, labels/arrows 1); position + nature 1. (b) 1 mark per use, max 2. Answers mixing Hindi/English words are accepted if the concept is correct.
17	5	(a) Ozone (O_3) is a molecule of three oxygen atoms; the ozone layer in the stratosphere absorbs harmful UV radiation. Depletion (mainly by CFCs) lets UV reach Earth → skin cancer, cataract, damage to plants/organisms. (b) Reduce: use less (switch off unused lights/fans). Reuse: use things again (refill bottles, reuse envelopes). Recycle: process waste into new products (paper, plastic, glass).	(a) What ozone is 1, why depletion matters 1. (b) 1 mark per R with a valid example. Answers continued on a later page must be read together. Hinglish acceptable.
	40	**Total**	